Lectures 7-8: Student Presentations of Foundational and Recent Papers

Session Format

Mini-Seminars

Structure: - Each student presents one selected paper - 8 minutes per presentation - Followed by peer discussion and Q&A (3 minutes) - Instructor moderates and connects to course themes

Objectives: - Expose class to broad range of soft matter topics - Practice critical reading skills - Develop scientific presentation abilities - Engage in scholarly discourse

Presentation Guidelines

What to Include

1. Paper Context
    - Authors, journal, year
    - Historical or scientific significance
    - Connection to course themes
2. Main Questions/Hypotheses
    - What problem does paper address?
    - What was known before?
    - What gap does it fill?
3. Key Methods
    - Experimental techniques OR
    - Theoretical approach OR
    - Computational methods
    - Briefly explain novel techniques
4. Main Results
    - Most important findings
    - Key figures/data
    - Quantitative results when relevant
5. Significance
    - Why does this matter?
    - Impact on field
    - Applications or implications
    - Connection to modern research
6. Critical Thinking
    - Assumptions made
    - Limitations
    - Future directions

Presentation Tips

- Visual: Use slides with clear figures
- Clarity: Explain jargon and technical terms
- Focus: Don't try to cover everything
- Engagement: Make eye contact, invite questions
- Time: Practice to stay within 8 minutes

Peer Discussion

Students should: - Ask clarifying questions - Pose critical questions - Make connections to other papers/lectures - Respectful and constructive

Example Questions: - "What are assumptions of that model?" - "How might experiment behave with different fluid?" - "How does this connect to [course topic]?"

Paper Topics (10 Options)

Paper 1: Brownian Motion and Atomic Theory

Citation: Einstein, A. (1905) "On the motion of small particles suspended in stationary liquids required by the molecular-kinetic theory of heat."

Key Points: - Explained jittery motion of pollen grains as molecular collisions - Provided direct evidence of atoms - Quantified diffusion - Related diffusion coefficient to particle size and temperature - Jean Perrin experimentally validated predictions (1908)

Why Foundational: - Cornerstone of soft matter physics - Introduced stochastic processes underlying colloidal science - Bridge between microscopic and macroscopic

Connection to Course: - Lecture 1: Thermal energy scale, mesoscopic structure - Underpins all subsequent colloidal physics

Paper 2: Entropic Forces and Liquid Crystal Ordering

Citation: Onsager, L. (1949) "The effects of shape on the interaction of colloidal particles" Ann. N.Y. Acad. Sci.

Key Points: - Theory of isotropic-nematic transition in rod-like colloids - Purely repulsive hard rods can spontaneously align at high concentration - Driven by entropy maximization - Increased orientational order frees up translational entropy - Drives nematic liquid crystal phase

Why Foundational: - Generalized how shape alone causes phase transitions - Laid groundwork for liquid crystal science - Polymer liquid crystals - Counter-intuitive: more order increases entropy!

Connection to Course: - Lecture 1: Entropy-driven behavior in soft matter - Shows complexity emerging from simple interactions

Paper 3: Polymer Dynamics (Reptation)

Citation: de Gennes, P.G. (1971) "Reptation of a Polymer Chain in the Presence of Fixed Obstacles" J. Chem. Phys.

Key Points: - Reptation model for polymer melts - Polymer moves snake-like through tube of confining neighbors - Explains dramatic viscosity increase in long entangled polymers - Specific scaling: diffusion ~ N^{-2} (N = polymer length) - Reptation length ~ N^{1}

Why Foundational: - Revolutionized polymer rheology - Molecular picture for viscoelasticity - Earned de Gennes Nobel Prize (1991) for soft matter universality

Connection to Course: - Lecture 5: Viscoelastic behavior of polymers - Lecture 6: Rheology, relaxation times

Paper 4: Auxetic Materials

Citation: Lakes, R.S. (1987) "Foam Structures with a Negative Poisson's Ratio" Science

Key Points: - Fabricated novel re-entrant foam - Expands laterally when stretched (Poisson's ratio = –0.6) - Counter-intuitive behavior from internal structure design - Hinged cell ribs create auxetic behavior - Improved stiffness and energy absorption

Why Foundational: - Opened field of mechanical metamaterials - Properties thought impossible in natural materials - Design-driven material properties

Connection to Course: - Soft matter engineering - Structure-property relationships - Mesoscopic design principles

Paper 5: Colloidal Glass Transition

Citation: Pusey, P.N. & van Megen, W. (1986) "Phase Behavior of Concentrated Suspensions of Nearly Hard Colloidal Spheres" Nature

Key Points: - Demonstrated glass transition in colloidal hard-spheres - Dramatic slowdown in particle motion at high volume fraction - Structural arrest using dynamic light scattering - Approaches random close packing (~58% volume fraction) - Analogous to molecular glass formation

Why Significant: - Established colloids as model for glassy dynamics - Bridge soft matter and condensed matter - Disorder-induced solidification is general phenomenon - Colloids allow real-space imaging of glass transition

Connection to Course: - Soft matter phase transitions - Role of volume fraction and packing - Non-equilibrium phenomena

Paper 6: Jamming Concept

Citation: Liu, A.J. & Nagel, S.R. (1998) "Jamming is Not Just Cool Any More" Nature

Key Points: - Unified disparate systems: granular media, glasses, foams - Proposed jamming phase diagram (density, load, temperature) - Above certain density + below certain temperature/shear $\rightarrow$ jammed (rigid) - "Fragile matter" rigidifies under stress - Collapses if stress direction changes

Why Important: - Interdisciplinary unification - How liquids solidify without crystalline order - Connected granular physics with phase transitions - Spawned new research field

Connection to Course: - Lecture 6: G' and G" behavior - Transition from fluid to solid - Shear thickening (related to paper 10)

Paper 7: Active Matter

Citation: Vicsek, T. et al. (1995) "Novel Type of Phase Transition in a System of Self-Driven Particles" Phys. Rev. Lett.

Key Points: - Simple model of flocking behavior - Point particles: constant speed + align with neighbors + noise - Continuous phase transition from disordered to collective motion - Non-zero average velocity when noise reduced or density increased - Now called "Vicsek model"

Why Significant: - Foundational in active matter field - Local alignment $\rightarrow$ macroscopic order - Applies to bird flocks, fish schools, self-propelled colloids - Spontaneous symmetry breaking in non-equilibrium systems

Connection to Course: - Non-equilibrium soft matter - Collective behavior - Pattern formation - Beyond thermal equilibrium

Paper 8: Mechanobiology

Citation: Engler, A.J. et al. (2006) "Matrix Elasticity Directs Stem Cell Lineage Specification" Cell

Key Points: - Stem cells sense substrate stiffness - Commit to different fates based on mechanical cues - Soft (~0.1-1 kPa, brain-like) $\rightarrow$ neurons - Intermediate (~10 kPa, muscle-like) $\rightarrow$ muscle cells - Stiff (~30-40 kPa, bone-like) $\rightarrow$ osteoblasts - Inhibiting cytoskeletal tension (myosin II) prevents effect

Why Groundbreaking: - Linked soft matter mechanics to biology - Tissue softness directs cell differentiation - Profound implications for tissue engineering - Understanding developmental biology

Connection to Course: - Lecture 6: Modulus as meaningful parameter (kPa range matters!) - Biological relevance of "softness" - Cells as mechanical sensors

Paper 9: Soft Robotics

Citation: Shepherd, R.F. et al. (2011) “Multigait soft robot” Proc. Natl. Acad. Sci.

Key Points: - Completely soft pneumatic robot from Whitesides’ group - Made of silicone elastomers - Could crawl and walk with different gaits - Switch between undulating snake-like and four-legged walking - Controlled by air inflation sequences - Could sustain impacts and squeeze through gaps

Why Notable: - Seminal demonstration of soft robotics - Functionalities hard for rigid robots - Safe human interaction - Morphing locomotion - Inspired wave of soft robotics research

Connection to Course: - Soft matter engineering - Rubber elasticity applications - Pneumatic networks - Search-and-rescue, medical devices

Paper 10: Shear Thickening & Dynamic Jamming

Citation: Waitukaitis, S.R. & Jaeger, H.M. (2012) “Impact-Activated Solidification of Dense Suspensions via Dynamic Jamming Fronts” Nature

Key Points: - Investigated cornstarch-water suspension (“oobleck”) - Turns solid-like under impact - High-speed imaging + force measurements - Rapidly moving object creates jamming front in suspension - Grains jam into temporary rigid network - Once front percolates, material solidifies locally

Why Significant: - Mechanism for discontinuous shear thickening - Links to transition from unjammed to jammed under stress - Explains everyday phenomena (running on oobleck!) - Novel impact-resistant materials - Shear-thickening fluids in protective gear

Connection to Course: - Lecture 6: Non-Newtonian rheology - Lecture 3: Instabilities and pattern formation - Dramatic material behavior change under stress

Schedule Overview

The final running order is set after paper selection. For a cohort of ten, each slot is 11 minutes: 8 minutes to present and 3 minutes for questions. This leaves 10 minutes for opening and synthesis within the two-hour session. Smaller cohorts use the same presentation limit with additional discussion time.

Instructor Role

During Presentations

- Time keeping
- Note connections to course material
- Prepare bridging comments

After Each Presentation

Connect to Course Concepts: - Example: “Einstein (1905) relates to Brownian motion and energy scales from Lecture 1” - Example: “Engler et al. (2006) ties into importance of modulus from Lecture 6”

Facilitate Discussion: - Encourage questions - Prompt deeper thinking - Ensure respectful dialogue

Wrap-Up

Course Synthesis: - Survey broad swath of soft matter - 100-year-old theories to cutting-edge research - Versatility and unifying principles of field - From fundamental physics to engineering applications - From molecular scale to macroscopic behavior

Learning Outcomes

By end of these sessions, students will:

1. Breadth of Soft Matter:
    - Appreciate historical development
    - Understand current research frontiers
    - See connections across sub-disciplines
2. Critical Reading:
    - Extract key points from scientific papers
    - Evaluate assumptions and limitations
    - Place work in broader context
3. Scientific Communication:
    - Present complex ideas clearly
    - Respond to questions professionally
    - Engage in scholarly discussion
4. Course Integration:
    - Connect papers to lecture themes
    - Synthesize knowledge across topics
    - Develop holistic understanding

Assessment Criteria

This final presentation contributes 60% of the course grade. The percentages below describe how the presentation mark itself is calculated.

Presentation (70%)

- Content Understanding (30%): Grasp of paper's main ideas
- Clarity of Explanation (20%): Clear, organized presentation
- Visual Aids (10%): Effective use of slides/figures
- Time Management (10%): Stay within 8 minutes

Discussion Participation (30%)

- Questions Asked (15%): Thoughtful questions to peers
- Answers Given (15%): Thoughtful responses to questions

Preparation Checklist

References

Full citations are provided in the *Student Presentation Topics* document.

Previous Lecture: *Lecture 6: Rheology* Course Home: *Course Overview* Paper Selection: *Detailed Paper Topics*

Metadata: Author:: Vatsal Sanjay Date published:: Jul 19, 2026 Date modified:: Jul 19, 2026